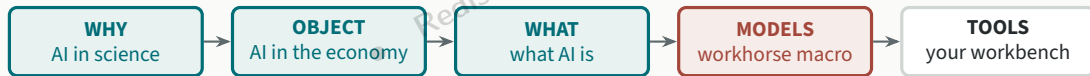


Session 1.4

Workhorse Macroeconomic Models

The models this course solves, the object each one asks for, and where grids stop working

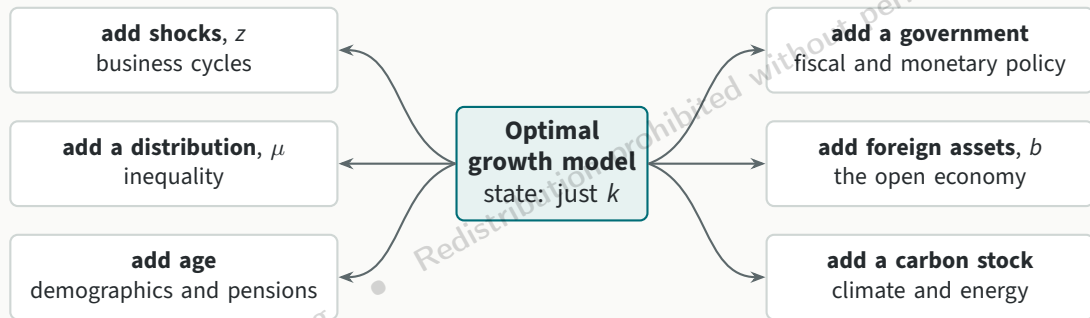


Zhigang Feng

Why This Deck, and Who It Is For

- ▶ The course is built for macroeconomists. The methods carry over to dynamic games in industrial organisation, asset pricing, and labour search.
- ▶ What AI will do for us: **(1)** solve the workhorse models efficiently and at scale; **(2)** turn unstructured data into measurement; **(3)** help us understand and write models, design algorithms, and implement code.
- ▶ **(4)** Verification, judgment and responsibility stay with the researcher.
- ▶ A word of warning: in this deck an **agent** is a household or a firm inside a model. That is the word's third meaning today, after the AI agent and the agent of contract theory.
- ▶ **For each model, ask: what is the state, what object must be computed, and what check would expose a wrong answer?**

One Model, Many Directions



- ▶ The trunk is one planner choosing consumption and capital. Each branch is a field of macroeconomics.
- ▶ **Every branch adds state variables, and more state means harder to compute.**

The Environment

$$\max_{\{c_t, k_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \log c_t \quad \text{subject to} \quad c_t + k_{t+1} = A k_t^\alpha, \quad k_0 \text{ given.}$$

- ▶ One good, one decision-maker. Output Ak^α is either eaten (c) or saved as tomorrow's capital (k').
- ▶ A is **total factor productivity** (TFP), α the capital share, β the discount factor.
- ▶ Log utility and full depreciation are chosen on purpose: this version has a closed-form solution.
- ▶ **State**: what you must know today to decide (k). **Control**: what you choose (k').
- ▶ **One calibration wherever this model returns (today's lab, Session 2)**: $\alpha = 0.33$, $\beta = 0.96$, $A = 1$.

Source: Brock & Mirman, "Optimal economic growth and uncertainty: the discounted case", *Journal of Economic Theory* (1972).

Why the Bellman Equation Can Be Trusted

$$V(k) = \max_{0 < k' < Ak^\alpha} \left\{ \log(Ak^\alpha - k') + \beta V(k') \right\} \equiv (TV)(k)$$

Result

What it guarantees

Principle of optimality	the sequence problem and the Bellman equation share a solution
Blackwell's conditions	monotonicity and discounting make T a contraction, modulus β
Contraction mapping	one fixed point V ; $V_{n+1} = TV_n$ reaches it from any V_0
Theorem of the maximum	V is continuous, and an optimal choice exists at every k
Concavity	V strictly concave, so the policy $k' = g(k)$ is a function
Benveniste–Scheinkman	V differentiable; the envelope condition gives the Euler equation

-
- **Two descriptions of one solution, a fixed point of T and the Euler equation: two ways to compute it.**

Source: Bellman (1957); Blackwell (1965); Benveniste & Scheinkman, *Econometrica* (1979); Stokey, Lucas & Prescott (1989), ch. 4.

Solving It, Approach 1: Iterate on the Value Function

Value function iteration

1. Grid $k_1 < \dots < k_N$; guess $V_0 \equiv 0$; tolerance ε .
2. For every k_i , the Bellman update:

$$V_{n+1}(k_i) = \max_{k_j < Ak_i^\alpha} \{ \log(Ak_i^\alpha - k_j) + \beta V_n(k_j) \},$$

and record the maximiser $g_{n+1}(k_i) = k_j$.

3. If $\max_i |V_{n+1}(k_i) - V_n(k_i)| < \varepsilon$, stop; otherwise $n \leftarrow n + 1$, go to 2.
4. Return V_{n+1} and g_{n+1} .

- Convergence is guaranteed: the error shrinks by β each step, 417 steps to 10^{-6} here.
- The policy settles long before the value. **Policy iteration** (Howard 1960) exploits this.
- **Cost per step:** $N \times N$ comparisons. **Session 3** replaces the grid with a network.

Source: Bellman (1957); Howard (1960). Iteration count at $\alpha = 0.33, \beta = 0.96, A = 1$.

Solving It, Approach 2: Solve the Equilibrium Conditions

Time iteration on the Euler equation

1. Grid $k_1 < \dots < k_N$; guess a policy g_0 , e.g.
 $g_0(k) = \frac{1}{2}Ak^\alpha$.

2. For every k_i , solve for k' :

$$\frac{1}{Ak_i^\alpha - k'} = \beta \frac{\alpha A k'^{\alpha-1}}{Ak'^\alpha - g_n(k')},$$

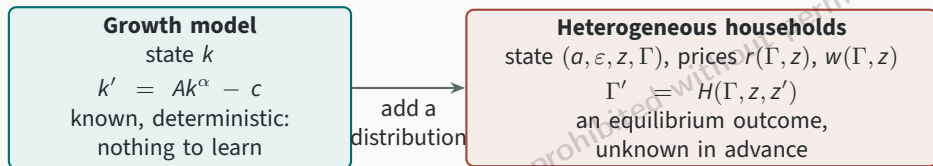
with g_n interpolated; set $g_{n+1}(k_i) = k'$.

3. Stop once $\max_i |g_{n+1}(k_i) - g_n(k_i)| < \varepsilon$.
4. Check the Euler residual on fresh k off the grid.

- The unknown is the policy g , never V .
- Variants: projection (Judd 1992) sets the residual to zero at nodes; endogenous grids (Carroll 2006) invert the equation, with no root-finding.
- **Euler alone admits explosive paths; transversality rules them out. Session 2 trains a network on the squared residual.**

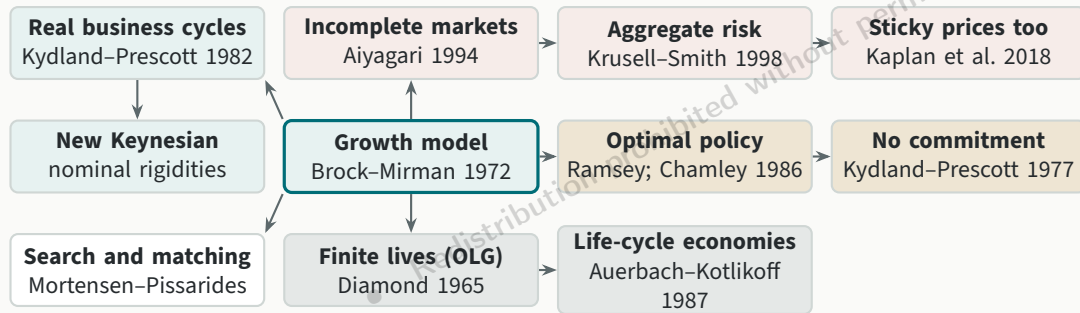
Source: Coleman (1990); Judd, *JET* (1992); Carroll, *Economics Letters* (2006).

When the Environment Is Unknown



- ▶ In reinforcement-learning terms the **environment** is the transition law and the reward.
- ▶ Γ is the distribution of households over wealth and productivity. Its law of motion H comes from everyone's choices, so it cannot be written down before the household problem is solved.
- ▶ Classical answer: guess H from a few moments (Krusell–Smith, later in this deck).
- ▶ Reinforcement learning: learn from simulated transitions (s, a, r, s') . H is known to the simulator, never to the learner, and expectations become sample averages.
- ▶ **Trivial in the growth model; the central difficulty once a distribution is a state.**

A Family Tree of Workhorse Models



- ▶ Teal: one representative household. Red: heterogeneity. Gold: policy. Grey: the life cycle.
- ▶ **Every branch now has a deep-learning frontier.** Sessions 2 to 4 walk the red, gold and grey ones.

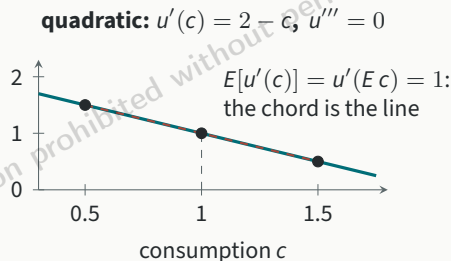
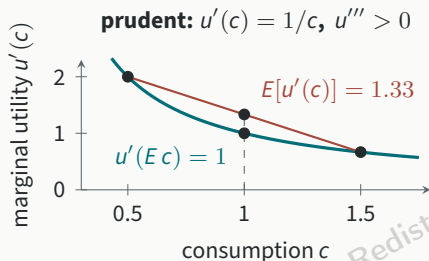
A Puzzle: Wealth Is Far More Unequal Than Income



- ▶ **Gini coefficient:** 0 if everyone holds the same, 1 if one household holds everything.
- ▶ The top 1% hold about a third of US wealth. A representative household has no distribution at all.
- ▶ The textbook incomplete-markets model gets well under half of the wealth Gini.
- ▶ **Every candidate fix (entrepreneurs, bequests, unequal returns) is a new state variable.**

Source: Survey of Consumer Finances 2007, in Díaz-Giménez, Glover & Ríos-Rull, *FRB Minneapolis Quarterly Review* (2011).

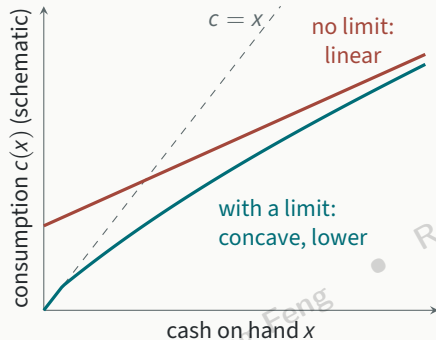
Why Households Self-Insure: Prudence, Not Risk Aversion



- ▶ Both are risk averse ($u'' < 0$); c_{t+1} is 0.5 or 1.5 with equal odds. In $u'(c_t) = \beta(1+r)E[u'(c_{t+1})]$, risk raises the right side only on the left.
- ▶ Left: saving rises. Right: saving is as under certainty (**certainty equivalence**).
- ▶ **Precautionary saving needs prudence ($u''' > 0$), not risk aversion alone.**

Source: Leland (1968); Sandmo (1970); Kimball, *Econometrica* (1990).

A Second Motive: The Borrowing Limit



- ▶ Quadratic utility, $u''' = 0$. Without a limit, $c(x)$ is linear and risk does not move it.
- ▶ With a limit, a bad draw tomorrow cannot be smoothed by borrowing. The household consumes less today, most of all when its cash on hand is low.
- ▶ **In Aiyagari both motives operate: prudence and the borrowing limit.**

Source: Zeldes, *Quarterly Journal of Economics* (1989); Carroll & Kimball (2001). Curves are schematic.

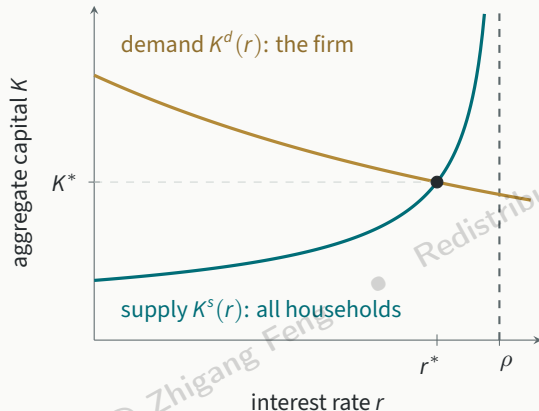
Aiyagari: The Household and the Firm

$$v(a, \varepsilon) = \max_{c, a'} \left\{ u(c) + \beta \sum_{\varepsilon'} \pi(\varepsilon' | \varepsilon) v(a', \varepsilon') \right\} \quad \text{s.t.} \quad c + a' = (1+r)a + w\varepsilon, \quad a' \geq -\underline{a}.$$
$$r = \alpha A (K/L)^{\alpha-1} - \delta, \quad w = (1 - \alpha) A (K/L)^{\alpha}.$$

- ▶ a : assets. ε : labour productivity, a Markov chain no one can insure. $\underline{a}$: the borrowing limit.
- ▶ A continuum of households, identical ex ante, different ex post: their histories differ.
- ▶ Each household's problem is a **Markov decision process**: state (a, ε) , action a' , reward $u(c)$, discount β , transition π .
- ▶ **The new object is the cross-section: a stationary distribution $\lambda(a, \varepsilon)$ of households.**

Source: Aiyagari, "Uninsured idiosyncratic risk and aggregate saving", *Quarterly Journal of Economics* (1994).

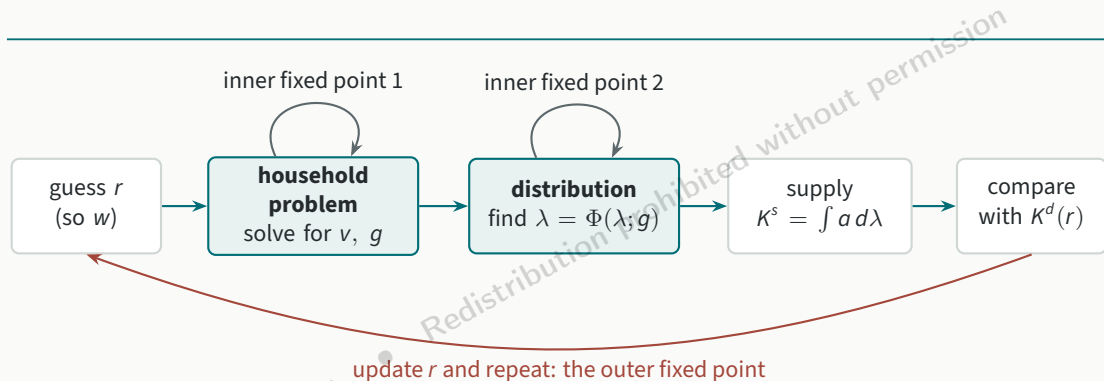
Equilibrium Is Where the Curves Cross



- ▶ $\rho = 1/\beta - 1$ is the rate of time preference. With full insurance, $r = \rho$.
- ▶ Savers facing risk would accumulate without bound as $r \rightarrow \rho$, so supply is finite only below it.
- ▶ Hence $r^* < \rho$, and capital sits above its complete-markets level.
- ▶ **That extra capital is not a free lunch. It is the cost of self-insurance.**

Source: Aiyagari (1994). A schematic: demand from the firm's first-order condition, supply asymptotic to $\rho = 1/\beta - 1$.

The Nested Fixed Point



- ▶ Every point on the supply curve contains two solved problems: a Bellman equation and an invariant distribution. Add a second price, or a tax, and the outer search becomes a system.
- ▶ **A small error at the outer level can hide a failure inside either inner problem.**

Krusell–Smith: The Distribution Starts to Move

Stationary world (Aiyagari). The distribution λ^* is an unknown we solve for *once*. Prices are numbers. Households forecast nothing.

add a TFP shock z

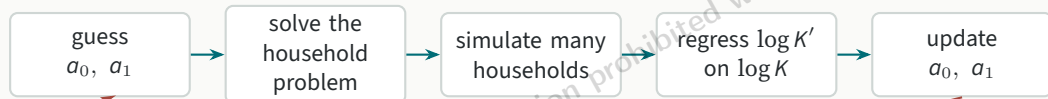
With aggregate shocks. The distribution Γ_t is a *state that moves every period*, $\Gamma' = H(\Gamma, z, z')$. Prices move with it. Households must forecast a distribution.

- ▶ Γ : the cross-sectional distribution of households over assets and productivity.
- ▶ Put wealth on 100 grid points and employment on 2. Γ is then a point in a 200-dimensional simplex.
- ▶ In Aiyagari we solved for one such point. Here we need a *function* of it.
- ▶ **The state is no longer a vector. It is a distribution.**

Source: Krusell & Smith, “Income and wealth heterogeneity in the macroeconomy”, *Journal of Political Economy* (1998).

The Krusell-Smith Bet

$$\log K' = a_0(z) + a_1(z) \log K$$

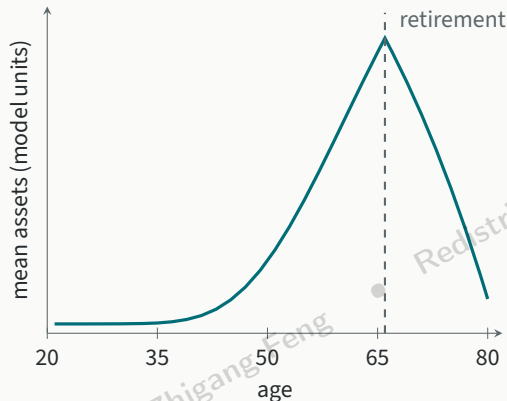


until the rule households use is the rule the simulation produces

- ▶ The bet: households forecast with one moment, mean capital K , not all of Γ . The fit is $R^2 > 0.9999$: **approximate aggregation**.
- ▶ R^2 **can flatter**. It scores one-step forecasts; errors that build up over many periods hide behind it (Den Haan 2010).

Source: Krusell & Smith (1998); Den Haan, *Journal of Economic Dynamics and Control* (2010).

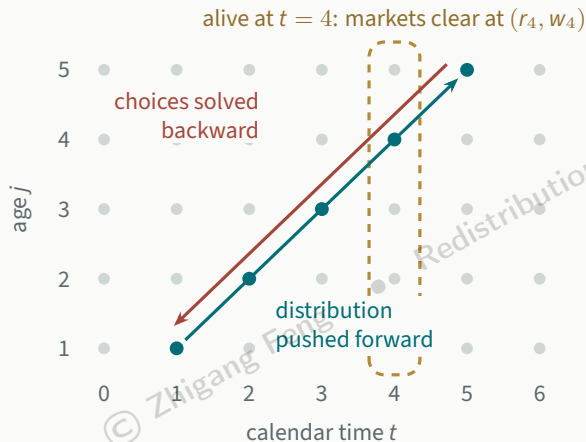
OLG: Age Becomes a State



- ▶ **Overlapping generations (OLG):** people live finite lives, and cohorts of different ages coexist.
- ▶ Sixty cohorts work to 65 and live to 80: they save for retirement, then run assets down.
- ▶ Finite lives help: solve backward from the last age, with no inner fixed point in values.
- ▶ **The state now carries age, so pensions, taxes and demographics can be studied.**

Source: Auerbach & Kotlikoff, *Dynamic Fiscal Policy* (1987). Profile: a 60-cohort model with a flat pension.

Transitions: Households Backward, Distributions Forward

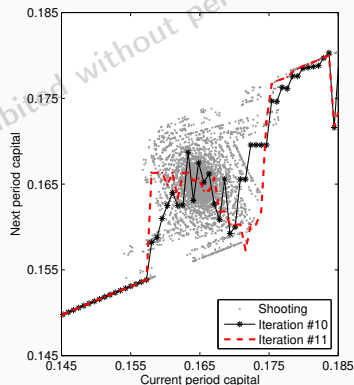


- ▶ A reform changes the whole price path $\{r_t, w_t\}$, not one number.
- ▶ Guess the path. Solve each cohort **backward** from its last age.
- ▶ Push the distribution **forward** from today's cross-section.
- ▶ Check market clearing in every period, update the path, repeat.
- ▶ **The unknown is a sequence of prices and distributions.**

Source: Auerbach & Kotlikoff (1987); Conesa & Krueger, "Social security reform with heterogeneous agents", *RED* (1999).

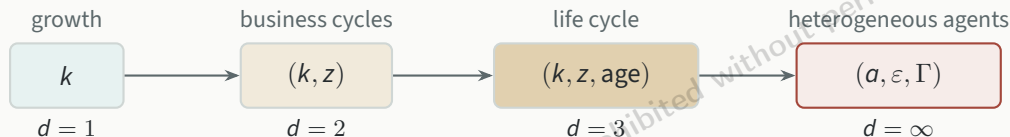
The Government Is a Player Too

- ▶ **Ramsey policy**: choose the whole tax path at date 0. Households' Euler equations become constraints.
- ▶ **Commitment adds a state**: a past promise, with a feasible range unknown in advance.
- ▶ **Time inconsistency** (Kydland–Prescott 1977): tomorrow's government will not keep today's plan.
- ▶ **Theory warns us**. A continuous equilibrium can fail to exist. A method that assumes a smooth policy then returns a confident, wrong answer.



Source: Santos, *Journal of Economic Theory* (2002). Right: his example computed numerically; iterations do not settle.

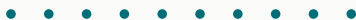
What Are We Solving For? The Objects Escalate



Model	Object to compute	Computational problem
Growth	value $V(k)$ and policy $g(k)$	one fixed point
Aiyagari	policies, a price r , a distribution λ	a nested fixed point
Krusell–Smith	policies and a law $\Gamma' = H(\Gamma, z, z')$	a fixed point in forecasting rules
OLG, Ramsey	policies by age, price paths, tax rules	backward and forward; a fixed point in rules

Why Grids Break

1 state: 10^2 points



2 states: 10^4 points



5 states: 10^{10} points

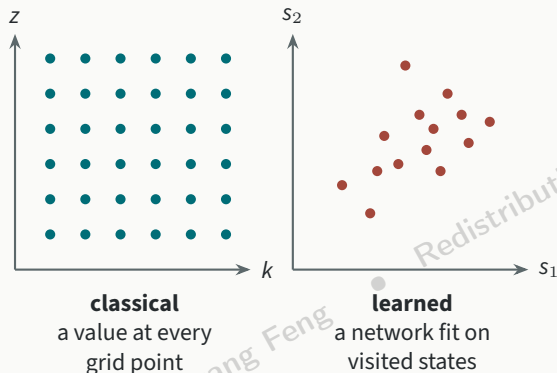


states d	grid points	one value table
2	10^4	0.08 MB
4	10^8	0.8 GB
6	10^{12}	8 TB

- **Curse one, the grid:** 100 points per state gives 100^d points.
- **Curse two, the expectation:** with 7 nodes per shock, 3 shocks cost $7^3 = 343$ terms at every grid point, every iteration.
- **A distribution as a state makes d effectively infinite.**

Source: Bellman (1957) coined “the curse of dimensionality”. Dots are a schematic sample, not the literal count.

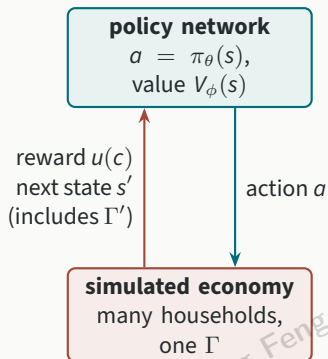
Where AI Enters (1): From Grids to Samples



- ▶ A neural network stands in for the policy or value function.
- ▶ All equilibrium conditions enter one loss, minimised by gradient descent.
- ▶ Training states come from simulating the model: **you pay only for the states the economy visits.**
- ▶ So Γ need not be collapsed to a few moments.

Source: Maliar, Maliar & Winant, *JME* (2021); Azinovic, Gaegauf & Scheidegger, *IER* (2022); Han, Yang & E, *QE* (2026); Fernández-Villaverde, Nuño & Perla, NBER WP 33117 (2024).

Where AI Enters (1), Continued: Reinforcement Learning



- **Critic**, target $\bar{\phi}$ held fixed:

$$\min_{\phi} \mathbb{E}[(u(c) + \beta V_{\bar{\phi}}(s') - V_{\phi}(s))^2]$$

- **Actor**, along simulated paths:

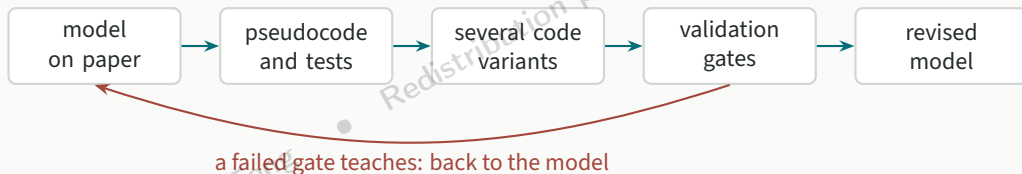
$$\max_{\theta} \mathbb{E}[u(c_{\theta}) + \beta V_{\phi}(s'_{\theta})]$$

- No transition law is written down: expectations are simulated averages.
- **Equilibrium: every household uses the trained rule. Sessions 3 and 4 build it.**

Source: Sutton & Barto, *Reinforcement Learning* (2018), ch. 13; Miao & Feng, *AI for Economists*, ch. 11.

Where AI Enters (2) and (3): Data, and a Collaborator

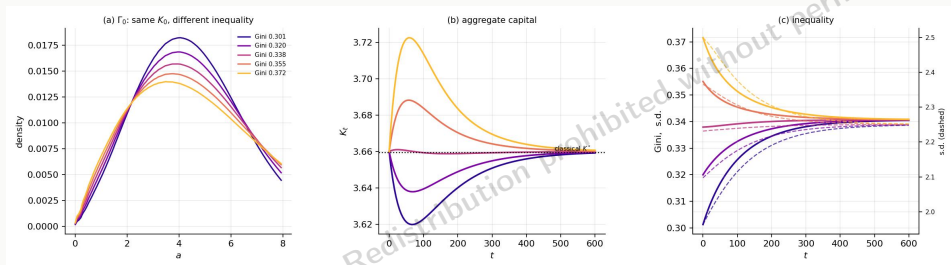
- ▶ **(2) Unstructured data become measurement.** Central-bank text gives a policy-stance index, night lights proxy for output, newspapers yield an uncertainty index.
- ▶ **(3) AI as a collaborator** helps you understand and write models, design algorithms, and implement code. Session 6 develops this loop.



- ▶ **AI writes variants quickly. You choose the state, the equilibrium concept, the benchmark, and what a failure means.**

Source: Hansen, McMahon & Prat (2018); Henderson, Storeygard & Weil (2012); Baker, Bloom & Davis (2016).

Research in Practice: Every Start Ends in the Same Place



- Five Aiyagari economies: the same $K_0 = 3.660$, Gini 0.301 to 0.372. All reach the classical $K^* = 3.6595$.
- A network trained on $\Gamma_{t+1} = \Phi(\Gamma_t)$ makes convergence a result, not an assumption.
- **Same K_0 , yet the distributions move for hundreds of periods. K alone misses it.**

Source: Feng, Han, Sargent & Zhu (2026), work in progress. Terminal K agrees to 3.5×10^{-8} .

What Stays With the Researcher

- ▶ A converged training loss certifies nothing. A solution is certified by a ladder of checks:
 - **Feasibility**: every choice respects the budget and borrowing constraints.
 - **Residuals on fresh draws**: unit-free Euler and Bellman errors, on states not used in training.
 - **A classical benchmark**: value function iteration on the low-dimensional case, where it still works.
 - **Theory**: policies are monotone, values concave, consumption smooth.
- ▶ The top rung is not computational: **your name on the result**.
- ▶ Today's lab climbs the first three rungs on the growth model, in under a minute.
- ▶ **Verification, judgment and responsibility stay with the researcher.**

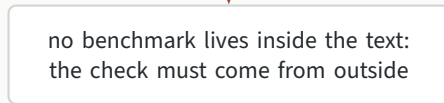
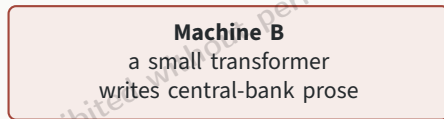
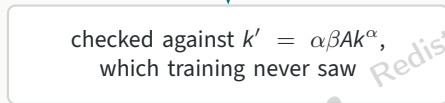
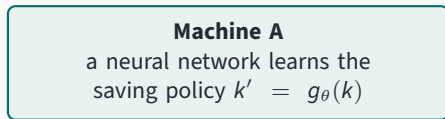
Source: Feng, "AI for Economics and the Economics of AI", working paper (2026), §3.3; Judd, Maliar & Maliar (2017).

Where Each Model Returns

Session	Object	What changes
1	Orientation	AI as tool, as object, and as an agent with incentives
2	Deep learning	Functions become trainable; equations become losses
3	Reinforcement learning	Sequential decisions are learned from interaction
4	Heterogeneous agents	Distributions and equilibrium become state variables
5	Text and LLMs	Language becomes data, measurement, and generation
6	RAG and agents	Retrieval, tools, traces, and governance in one workflow

- ▶ The growth model returns in Sessions 2 and 3; Aiyagari, Krusell–Smith, OLG and optimal taxation in Session 4.
- ▶ **Next: a first lab that solves the growth model with a network, and checks it. Deck 1.5 sets up the workbench it runs on.**

Lab: Two Machines, One Environment



- ▶ Machine A minimises the squared Euler residual of Approach 2.
- ▶ Machine B learned the *look* of FOMC statements, not facts: no residual to compute.
- ▶ **Same technology, two epistemic positions: one output you can verify, one you cannot.**